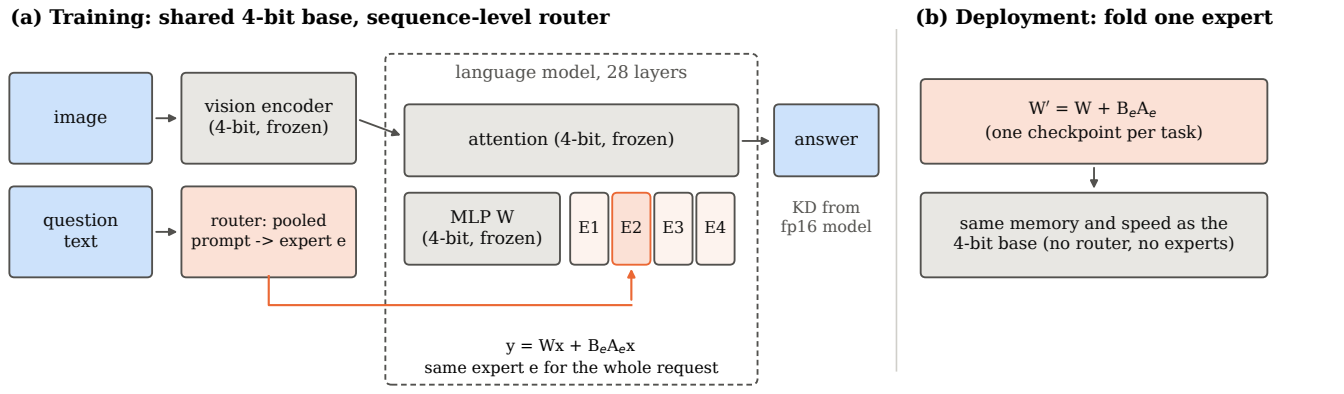
(a) Training: shared 4-bit base, sequence-level router
image
vision encoder
(4-bit, frozen)
question
text
router: pooled
prompt -> expert e
language model, 28 layers
attention (4-bit, frozen)
MLP W
(4-bit, frozen)
E1
E2
E3
E4
answer
KD from
fp16 model
y = Wx + BeAex
same expert e for the whole request
(b) Deployment: fold one expert
W′ = W + BeAe
(one checkpoint per task)
same memory and speed as the
4-bit base (no router, no experts)